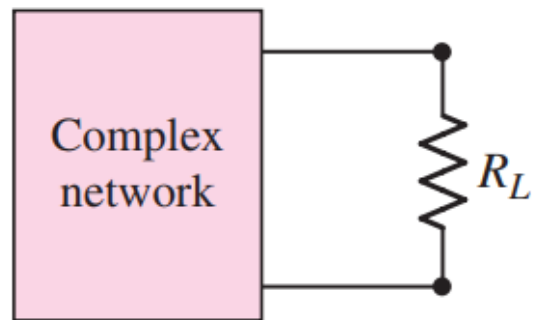
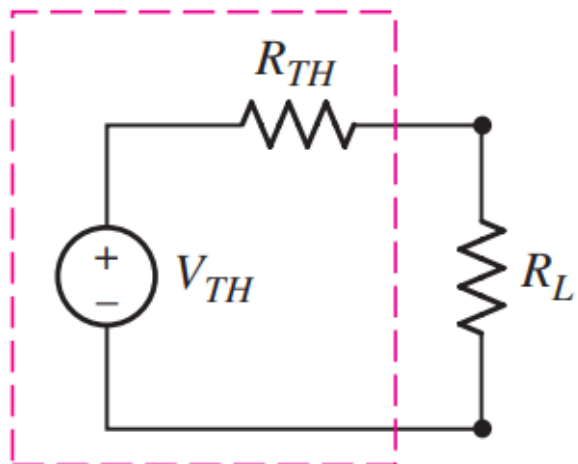
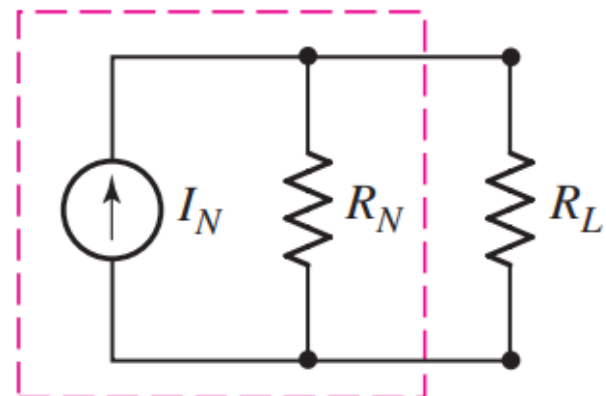




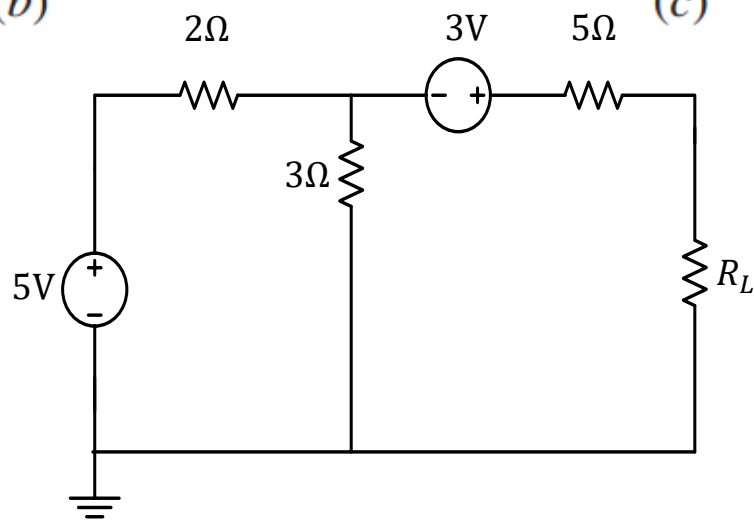
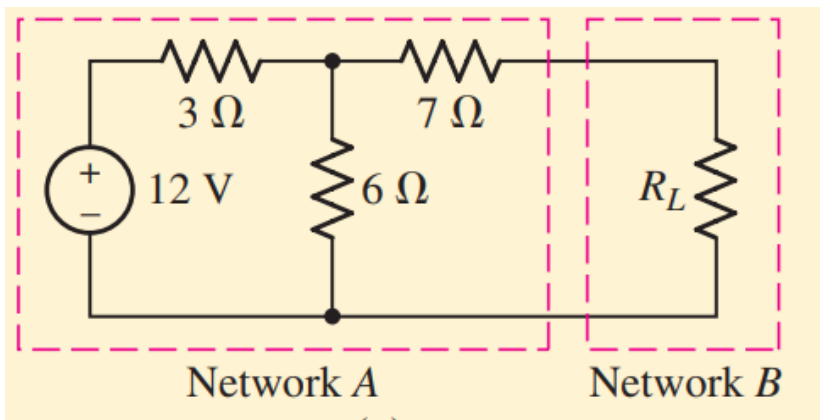
(a)



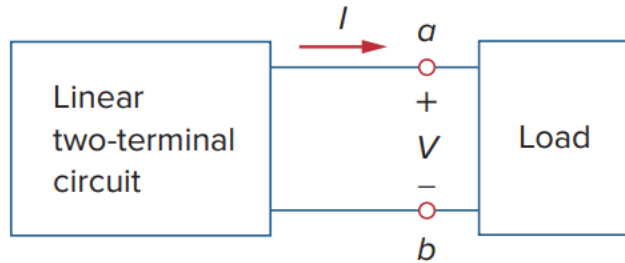
(b)



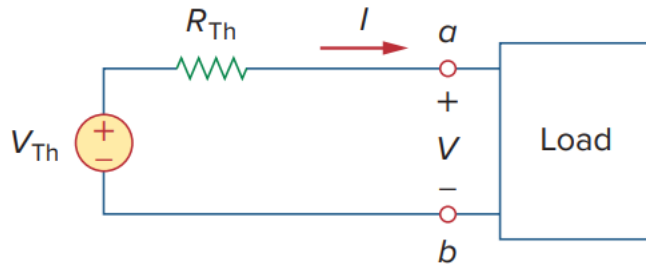
(c)

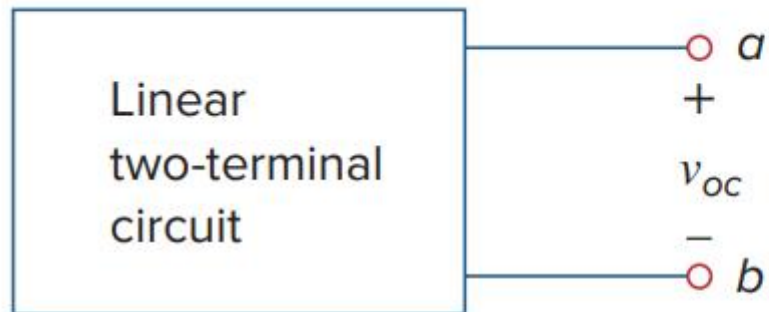


Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} , where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.



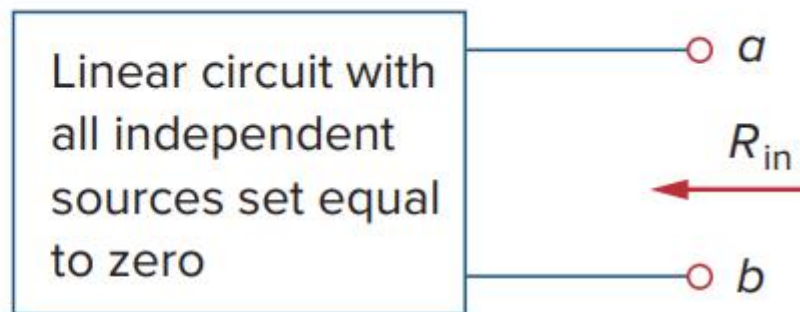
(a)





$$V_{Th} = v_{oc}$$

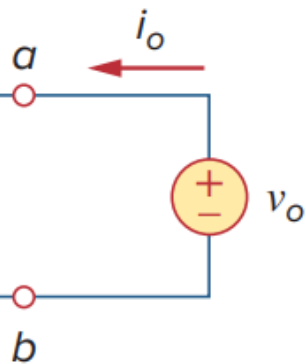
(a)



$$R_{Th} = R_{in}$$

(b)

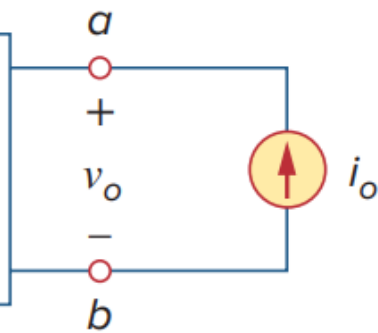
Circuit with
all independent
sources set equal
to zero



$$R_{Th} = \frac{v_o}{i_o}$$

(a)

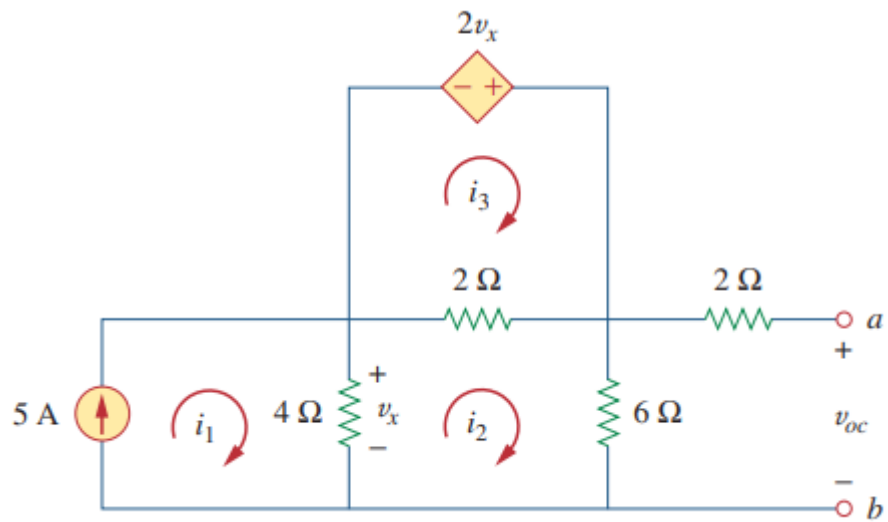
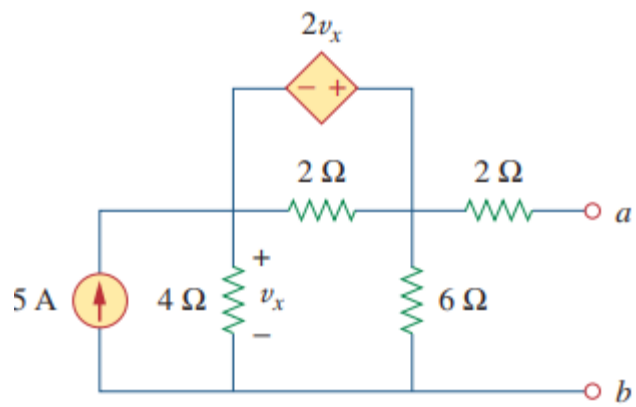
Circuit with
all independent
sources set equal
to zero

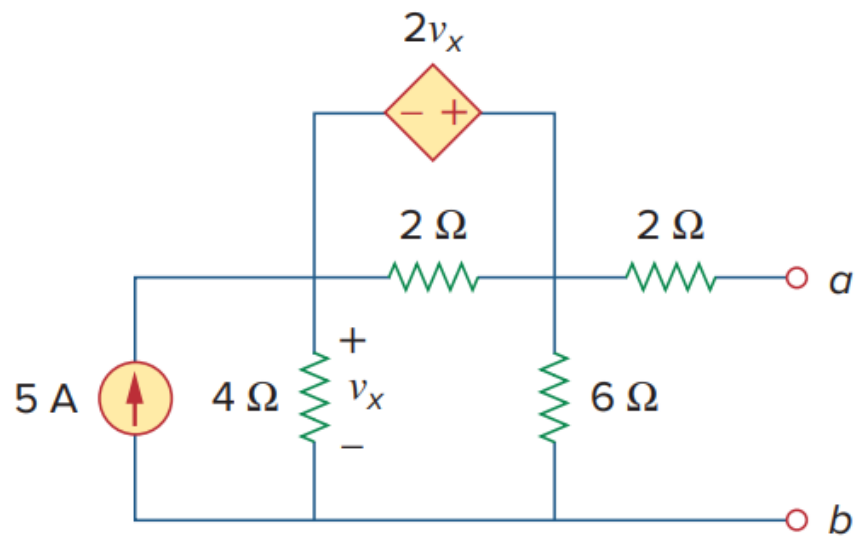


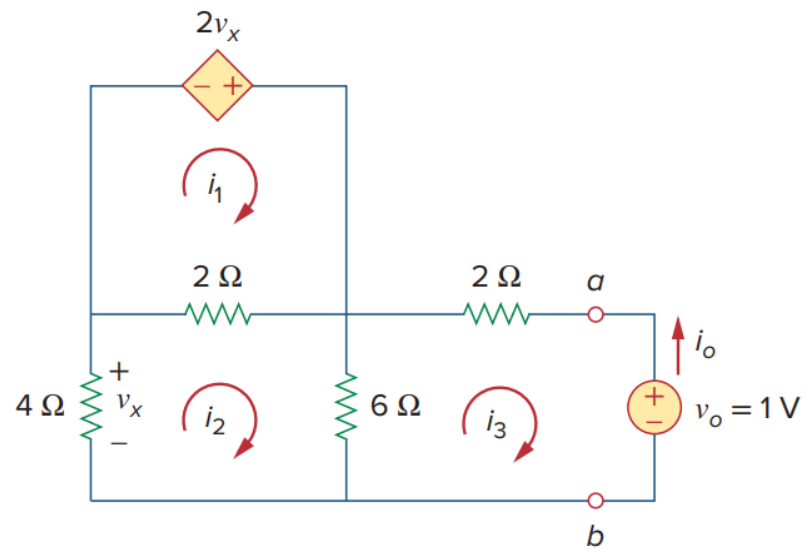
$$R_{Th} = \frac{v_o}{i_o}$$

A Statement of Thévenin's Theorem

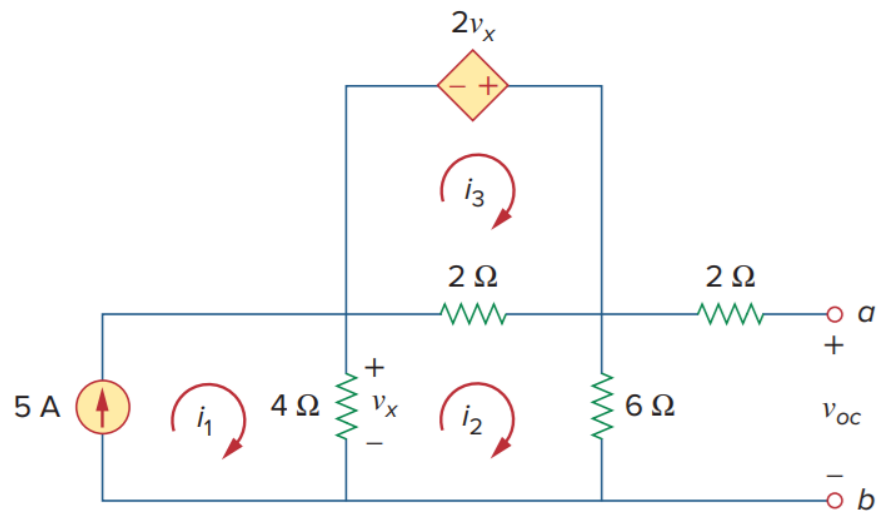
1. **Given any linear circuit, rearrange it in the form of two networks, A and B , connected by two wires.** Network A is the network to be simplified; B will be left untouched.
2. **Disconnect network B .** Define a voltage v_{oc} as the voltage now appearing across the terminals of network A .
3. **Turn off or “zero out” every independent source in network A to form an inactive network.** Leave dependent sources unchanged.
4. **Connect an independent voltage source with value v_{oc} in series with the inactive network.** Do not complete the circuit; leave the two terminals disconnected.
5. **Connect network B to the terminals of the new network A .** All currents and voltages in B will remain unchanged.







(a)



(b)

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

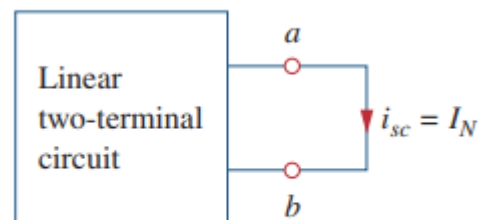
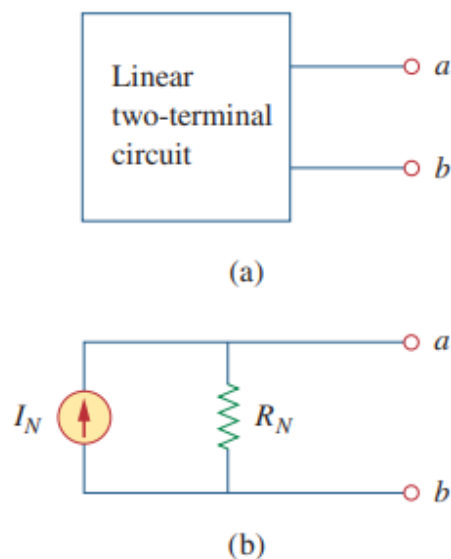
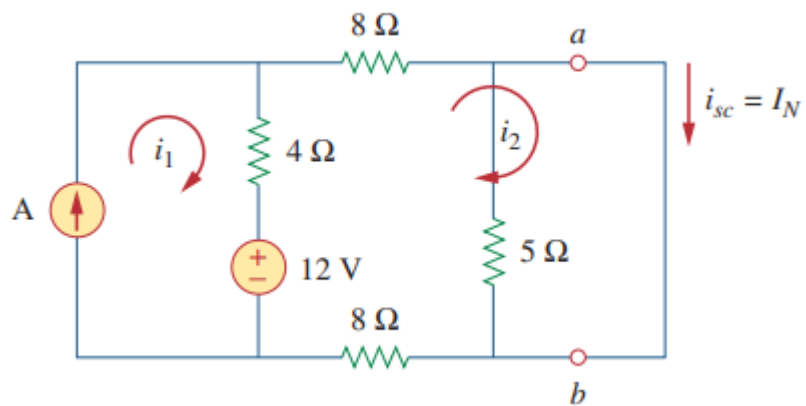
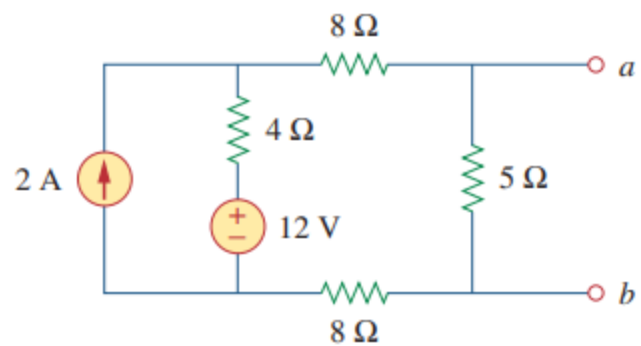


Figure 4.38
Finding Norton current I_N .



MAXIMUM POWER TRANSFER

In many practical situations, a circuit is designed to provide power to a load. There are applications in areas such as communications where it is desirable to maximize the power delivered to a load.

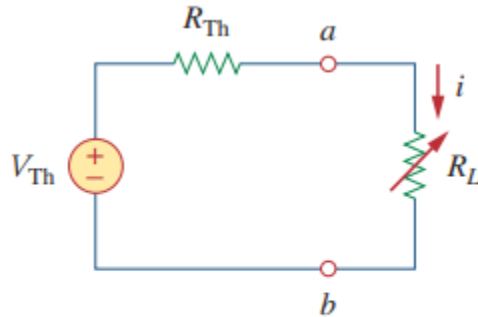


Figure 4.48

The circuit used for maximum power transfer.

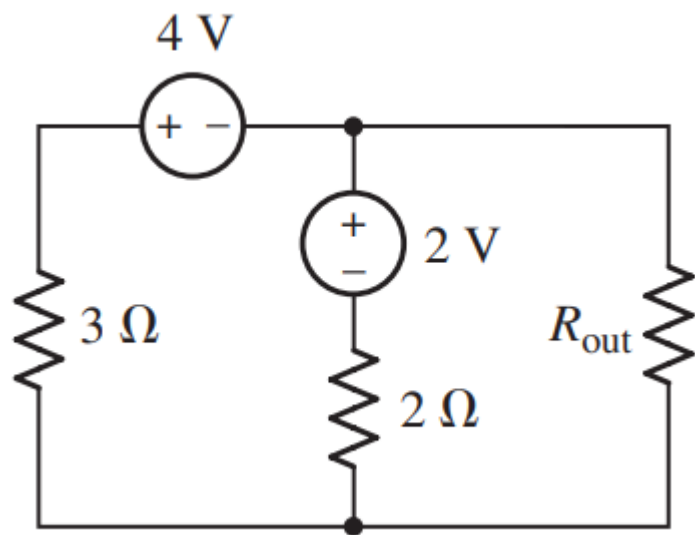
$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

$$\begin{aligned} \frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \end{aligned}$$

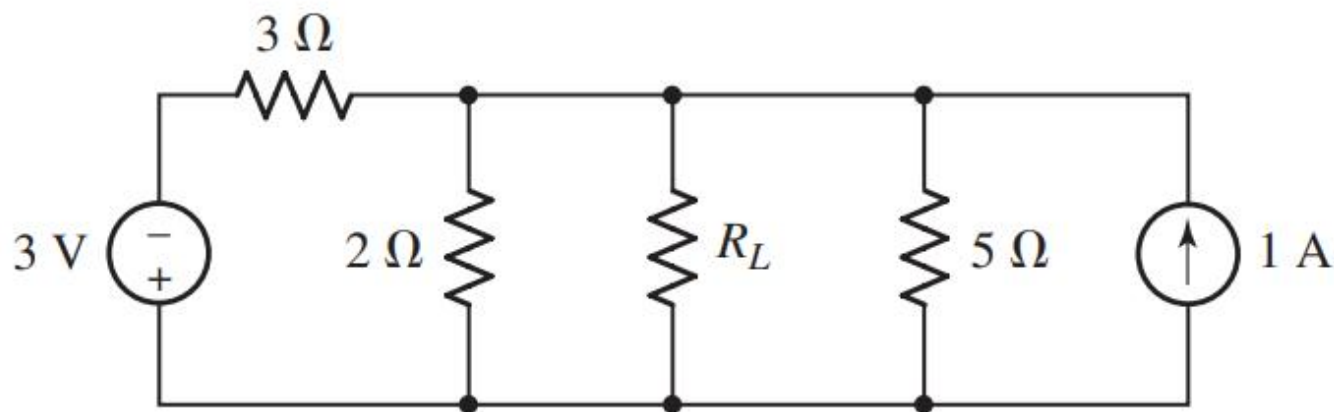
$$R_L = R_{Th}$$

$$p_{\max} = \frac{V_{Th}^2}{4R_{Th}}$$

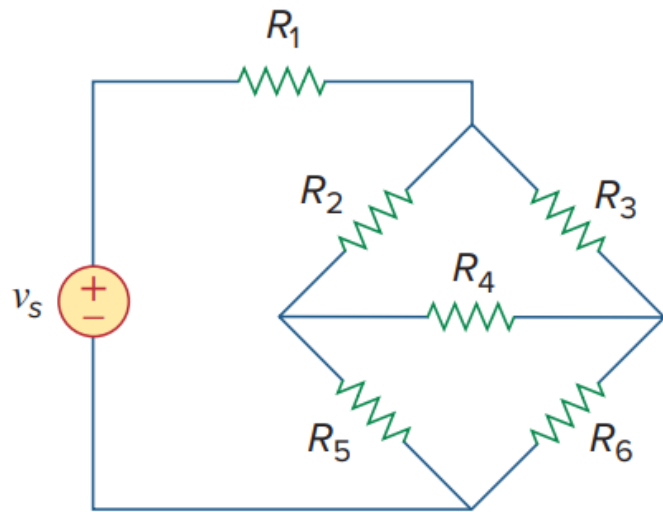
47. For the circuit drawn in Fig. 5.88, (a) determine the Thévenin equivalent connected to R_{out} . (b) Choose R_{out} such that maximum power is delivered to it.



50. For the circuit of Fig. 5.90, what value of R_L will ensure it absorbs the maximum possible amount of power?



■ **FIGURE 5.90**



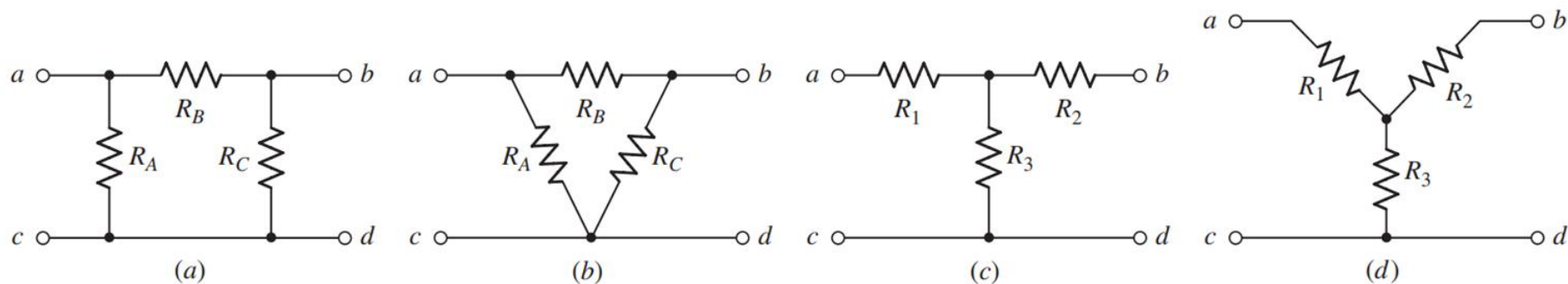


FIGURE 5.44 (a) Π network consisting of three resistors and three unique connections. (b) Same network drawn as a Δ network. (c) A T network consisting of three resistors. (d) Same network drawn as a Y network.

$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C}$$

$$R_2 = \frac{R_B R_C}{R_A + R_B + R_C}$$

$$R_3 = \frac{R_C R_A}{R_A + R_B + R_C}$$

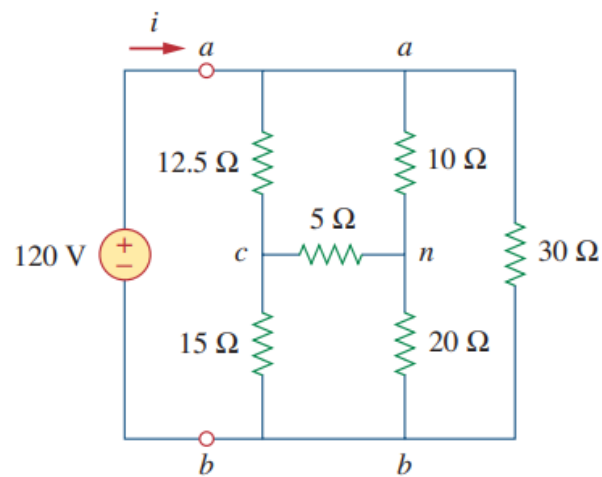
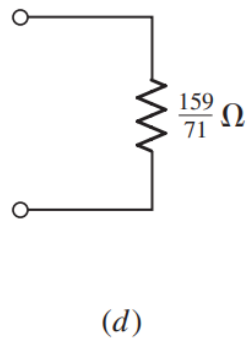
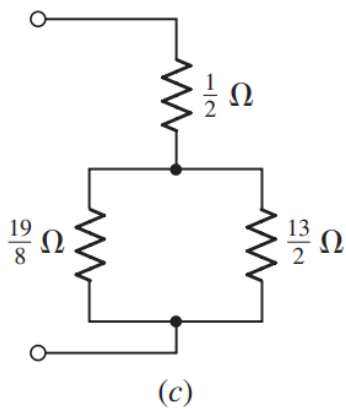
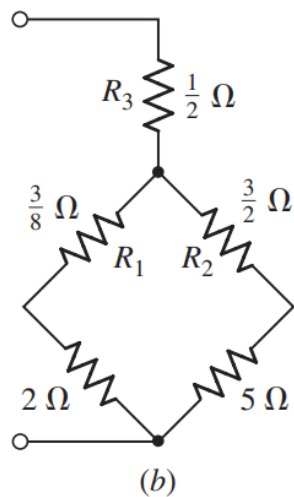
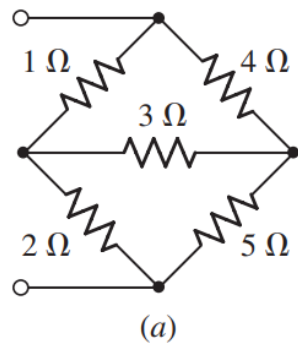


Figure 2.52
For Example 2.15.